

TYING THE (DATA) KNOT: LOVE, LIFE & LIKES

Predicting Relationship Connection
Outcomes Using Machine Learning



PROJECT OVERVIEW

THIS PROJECT AIMS TO PREDICT WHETHER A DATING APP INTERACTION WILL LEAD TO A SUCCESSFUL CONNECTION.

Using a dataset of 50,000 dating profiles, we developed an end-to-end machine learning pipeline involving:

- Data preprocessing
- Feature engineering
- Feature selection
- Model training
- Explainable AI
- Causal inference



PROBLEM STATEMENT

Dating platforms rely heavily on recommendation systems.

However:

- User behaviour is highly complex
- Many variables influence relationships
- Correlation does not imply causation

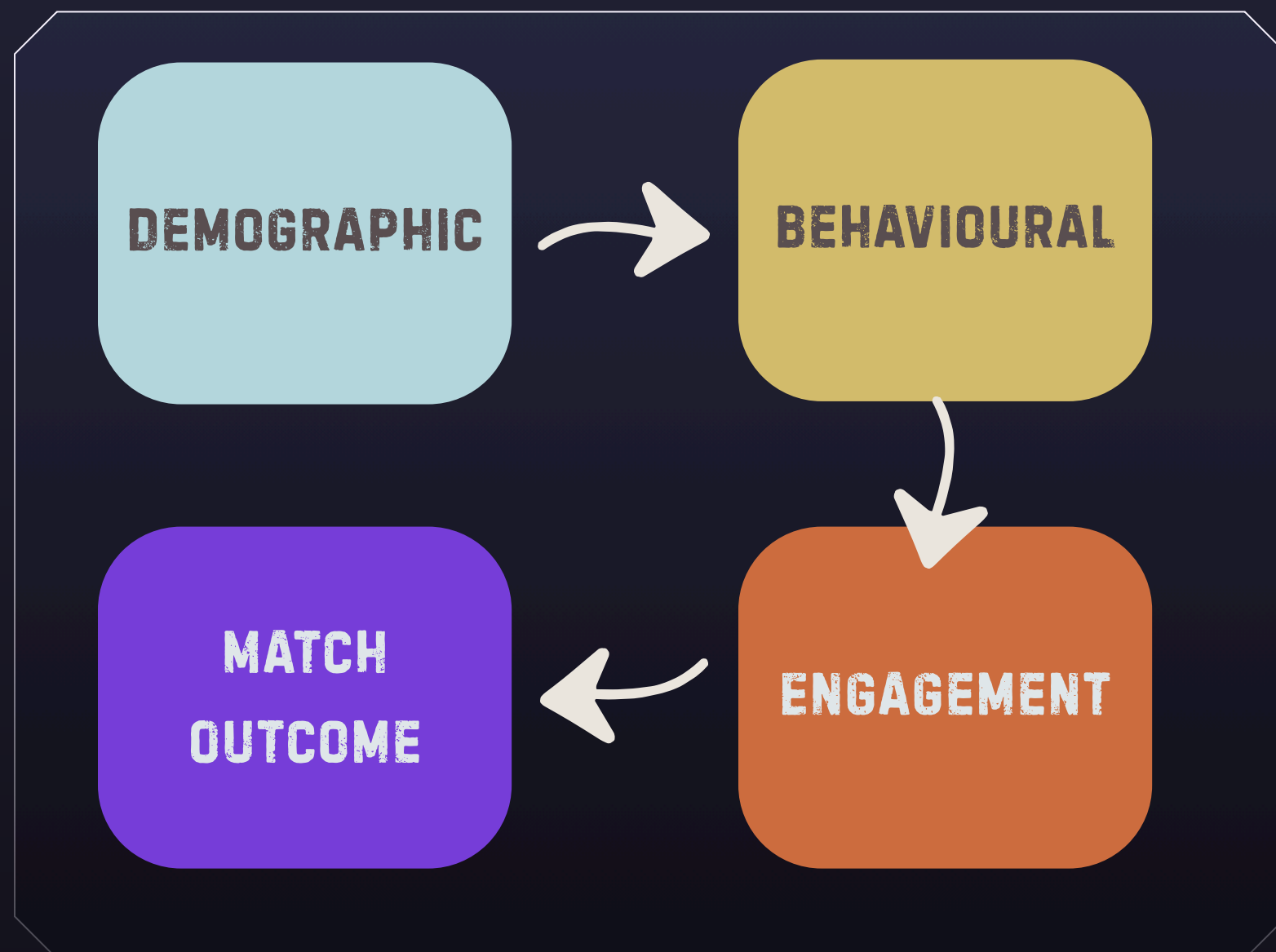


KEY QUESTION:

Can machine learning accurately predict successful relationship outcomes using behavioural and demographic information?



DATASET OVERVIEW



Dataset Source:

Kaggle Dating App Behavior Dataset

Dataset Characteristics:

- 50,000 records
- 25 raw features
- Demographic information
- Behavioural patterns
- Match outcomes
-

Examples of Features:

- Age
- Gender
- Education
- Income
- Likes received
- Mutual matches
- Swipe ratio
- Message count

TARGET ENGINEERING

Original Dataset:

10 relationship outcome categories

Examples:

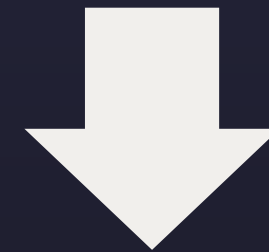
- Mutual Match
- Long-Term Relationship
- Ghosted
- Catfished
- No Response

Converted into Binary Classification:

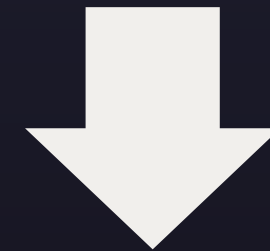
Positive Outcome → Successful Connection

Negative Outcome → Unsuccessful Connection

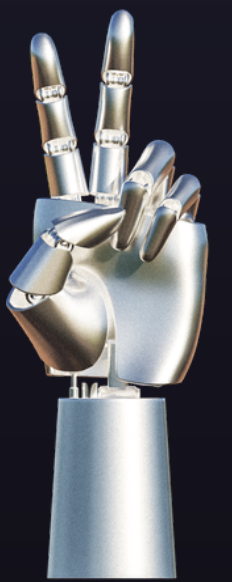
10 CLASSES



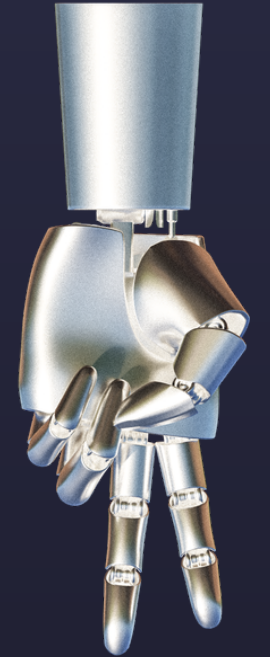
BINARIZATION



CONNECTION = 1
NO CONNECTION = 0

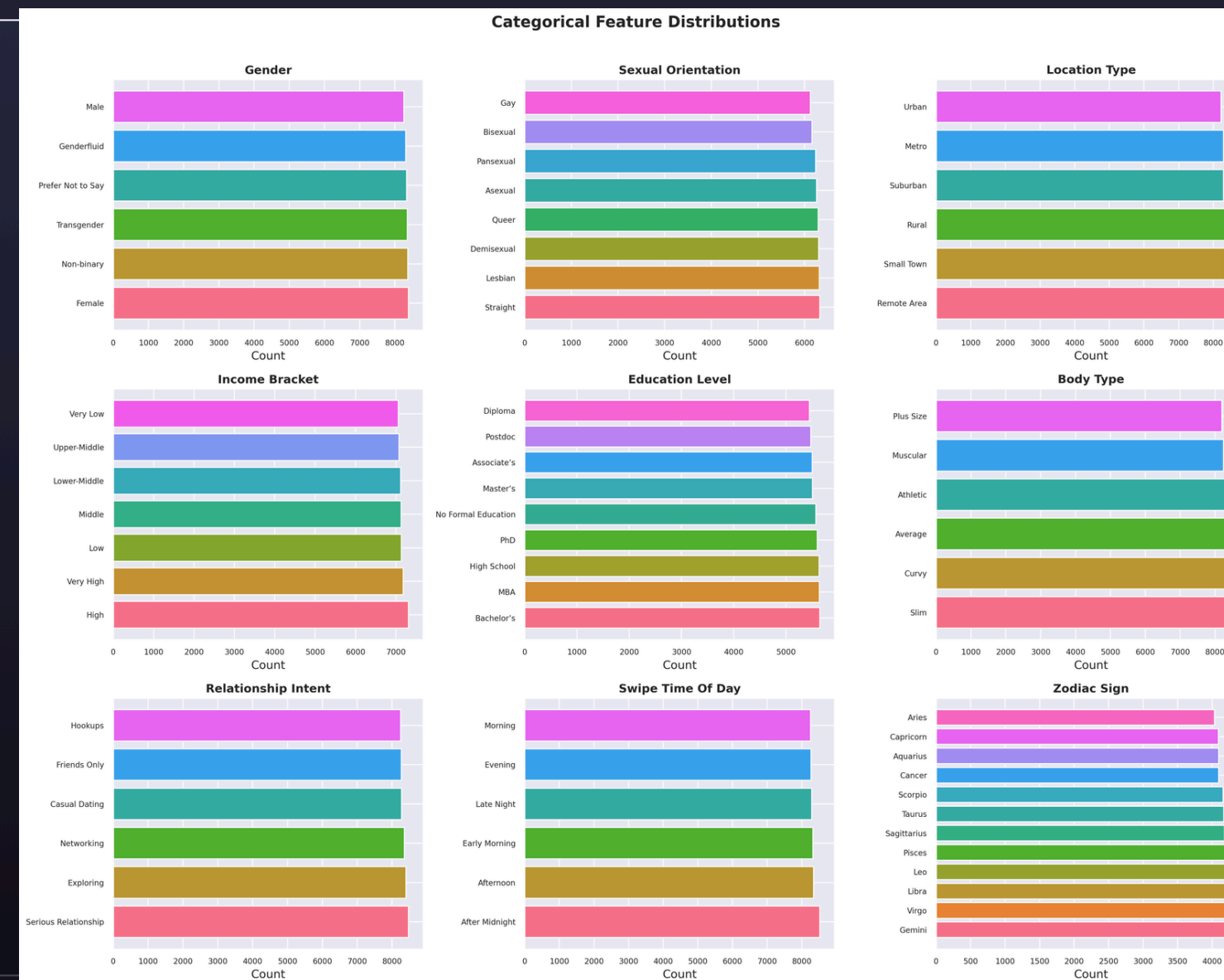


EXPLORATORY DATA ANALYSIS (EDA)



Key Findings:

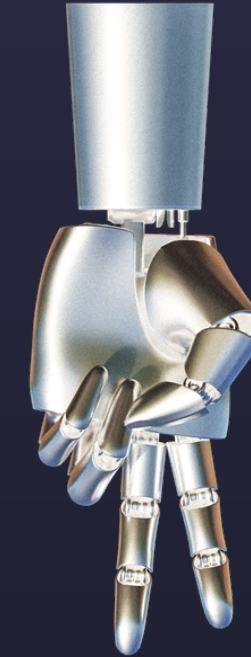
- No missing values
- No duplicate records
- Balanced synthetic dataset



"WHY EDA MATTERS"

- DATA QUALITY VERIFICATION
- DISTRIBUTION ANALYSIS
- FEATURE BEHAVIOUR UNDERSTANDING

NUMERICAL FEATURE DISTRIBUTIONS



Observations:

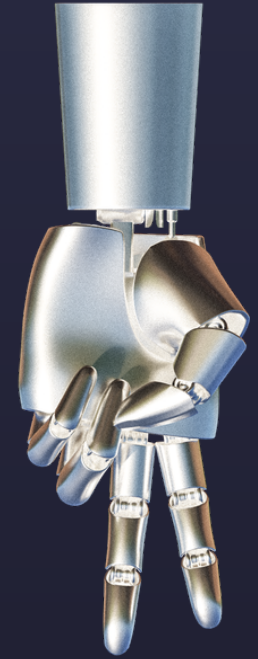
- Age follows normal distribution
- Height approximately symmetric
- Activity metrics highly skewed
- Likes and messages contain heavy tails

Implication:

Standard scaling unsuitable.
RobustScaler selected.



OUTLIER ANALYSIS



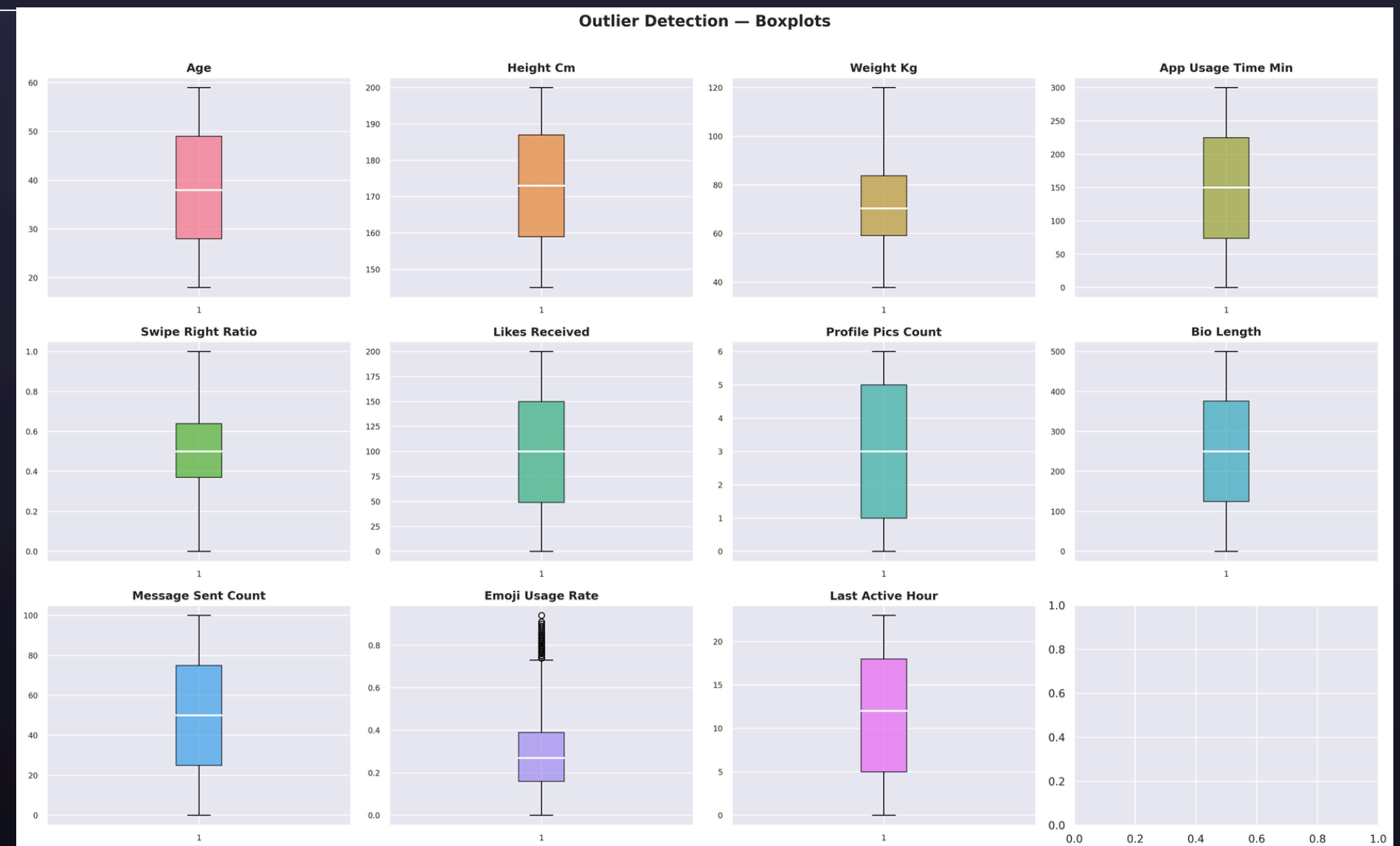
Power users produce extreme values in:

- Likes received
- Mutual matches
- Messages sent

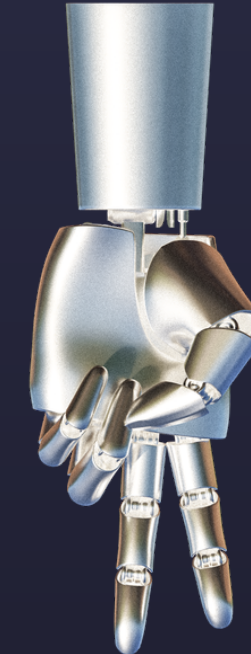
These outliers may distort model behaviour.

Solution:

Isolation Forest OOD Detection



COMPLETE ML PIPELINE



DATA PREPROCESSING PIPELINE

Applied:

- Target Binarization
- Ordinal Encoding
- One-Hot Encoding
- Multi-Hot Encoding
- Robust Scaling

Generated:

67 Machine Learning Features

SMARTART PIPELINE:

Raw Data



Encoding



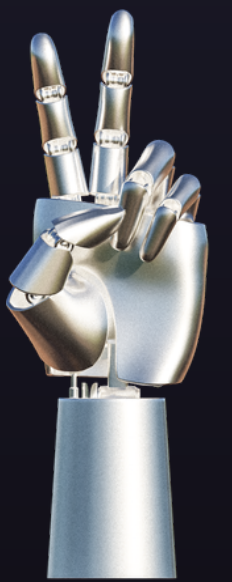
Scaling



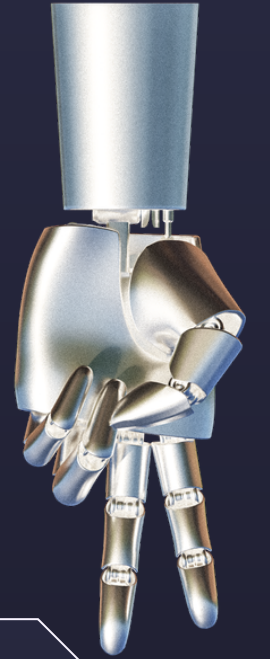
Feature Engineering



ML Features



FEATURE ENGINEERING



Created behavioural interaction features:

1. *Popularity Density*

LIKES RECEIVED / USAGE DURATION

2. *Bio Message Interaction*

BIO LENGTH × MESSAGE COUNT

3. *Selective Emoji Swiper*

EMOJI USAGE × SWIPE BEHAVIOUR

Purpose:

Capture hidden behavioural patterns.

OUT-OF-DISTRIBUTION DETECTION

Isolation Forest was deployed before classification.

Purpose:

- Detect abnormal profiles
- Reject anomalous inputs
- Improve reliability

Profiles exceeding anomaly threshold are excluded from prediction.

USER PROFILE



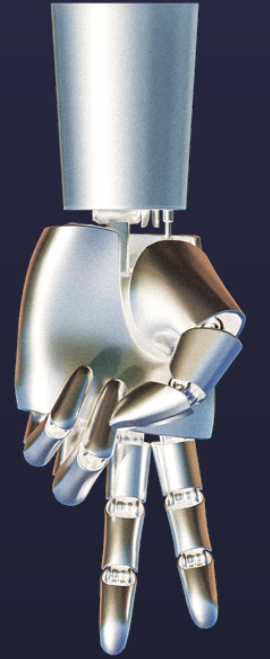
ISOLATION FOREST



NORMAL → MODEL
ANOMALY → REJECT



FEATURE SELECTION STRATEGY



Three complementary methods used:

- ANOVA F-Test
- Mutual Information
- Boruta

Final Features:

67 features retained from ensemble voting strategy

Benefits:

- Reduce noise
- Improve efficiency
- Retain useful information

METHOD	CAPTURES
ANOVA	LINEAR RELATIONSHIPS
MUTUAL INFORMATION	NON-LINEAR RELATIONSHIPS
BORUTA	TREE-BASED IMPORTANCE



MODELS EVALUATED

14 MODELS TESTED

Why Advanced Models?

- FT-Transformer
Attention between features
- SAINT
Attention between features and samples
- NODE
Differentiable decision trees
- TabPFN
Foundation model for tabular data

TRADITIONAL:

- LOGISTIC REGRESSION
- KNN
- DECISION TREE
- RANDOM FOREST

ENSEMBLE:

- XGBOOST
- LIGHTGBM
- CATBOOST

DEEP LEARNING:

- FT-TRANSFORMER
- SAINT
- NODE

ADVANCED:

- TABPFN
- SVM BAGGING

TRAINING STRATEGY

Training Process:

- 80/20 Train-Test Split
- Stratified Sampling
- SMOTE Balancing
- RandomizedSearchCV
- 10-fold Cross Validation

GOAL:
OBTAIN FAIR AND ROBUST EVALUATION.

LEAKAGE PREVENTION

- **TRAIN/TEST SPLIT FIRST**
- **SMOTE APPLIED ONLY ON TRAINING DATA**
- **CROSS-VALIDATION INSIDE TRAINING PIPELINE**

PERFORMANCE COMPARISON

Comparison based on:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Classifier	Test	Precision	Recall	F1-Score	ROC-AUC	Fit Time (s)
Logistic	53.03%	40.05%	36.88%	38.40%	0.5033	0.27
K-Nearest	43.28%	39.62%	81.79%	53.38%	0.5055	0.02
Decision	60.03%	38.84%	1.18%	2.30%	0.4993	0.45
Random	52.25%	39.34%	37.43%	38.36%	0.4992	4.38
XGBoost	54.18%	40.03%	30.93%	34.90%	0.5052	2.39
LightGBM	57.81%	39.72%	12.12%	18.57%	0.5112	367.54
CatBoost	56.80%	40.18%	18.04%	24.90%	0.5003	12.12
SVM Bagging	60.30%	0.00%	0.00%	0.00%	0.5143	1983.47
Multi-Layer	60.30%	0.00%	0.00%	0.00%	0.5	12.4
Graph	60.30%	0.00%	0.00%	0.00%	0.5	34.5
SCARF	60.30%	0.00%	0.00%	0.00%	0.5	18.20 (pre)
Opacus DP-	60.30%	0.00%	0.00%	0.00%	0.5	45.1
TabPFN	59.90%	40.74%	2.22%	4.20%	0.5005	1.85
Label	60.05%	47.80%	14.20%	21.89%	0.5052	15.4
TabNet-style	60.15%	48.50%	9.80%	16.31%	0.5031	22.8
Cosine KNN	47.22%	39.34%	60.81%	47.77%	0.4994	0.02

KEY OBSERVATION :
ALL MODELS
CONVERGE NEAR 0.50
ROC-AUC.

CHAMPION MODEL : LIGHTGBM

Why LightGBM Won

- ✓ Highest practical ROC-AUC
- ✓ Native SHAP compatibility
- ✓ Fast deployment
- ✓ Handles mixed feature types

Performance:

- Accuracy: 57.81%
- ROC-AUC: 0.5112



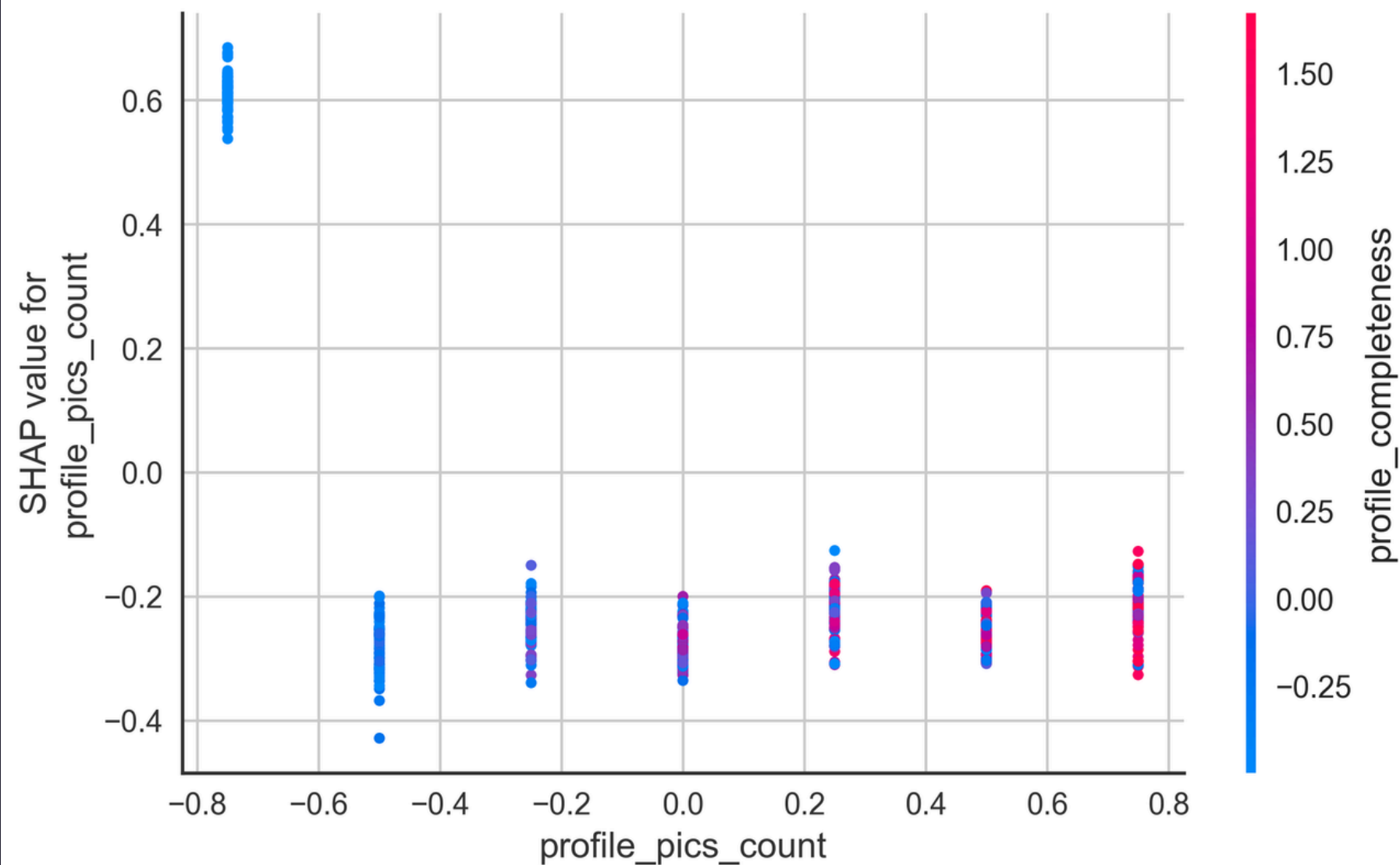
ROC-AUC = 0.5112

Key Insight

- SHAP values remain clustered near zero.
- No strong interaction effects observed.
- Feature contributions are weak.

MODEL INTERPRETABILITY WITH SHAP

SHAP Interaction Analysis: profile_pics_count × profile_completeness



SHAP explains:

- Global feature importance
- Individual predictions
- Feature impact direction

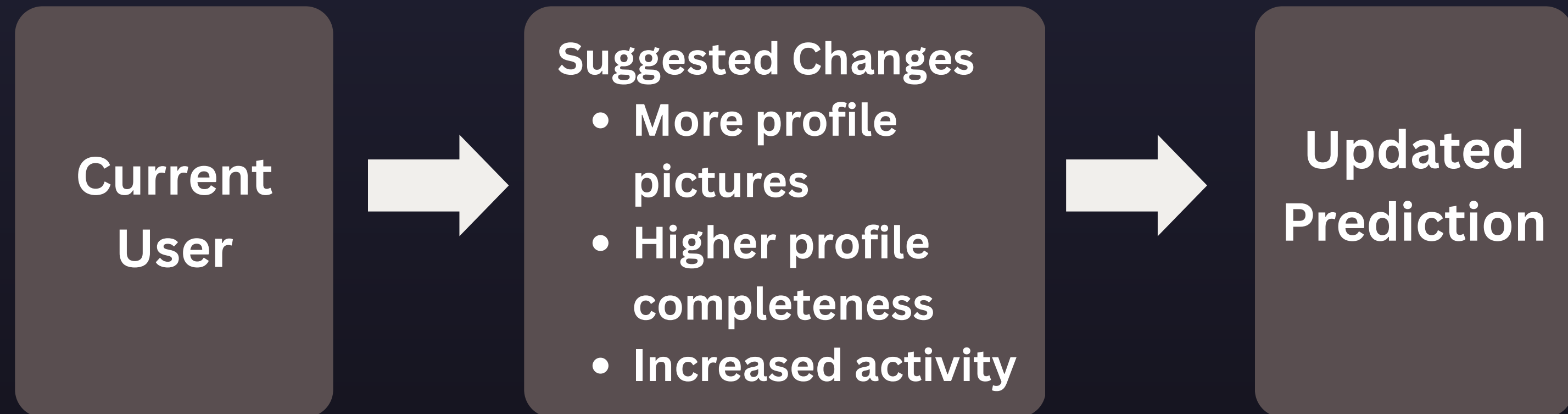
Benefits:

- Transparency
- Trustworthiness
- Explainability



COUNTERFACTUAL EXPLANATIONS (DICE)

WHAT SHOULD A USER CHANGE?



DOUBLE MACHINE LEARNING (DML)

Why DML?

Prediction answers:

What is likely to happen?

DML answers:

What happens if behaviour changes?

Result:

ATE = 0.0104

Interpretation:

Behavioural interventions produce only a very small improvement in connection probability.

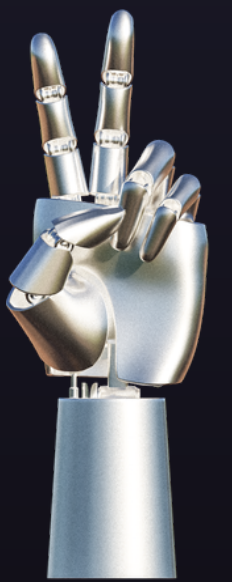
TREATMENT



DML

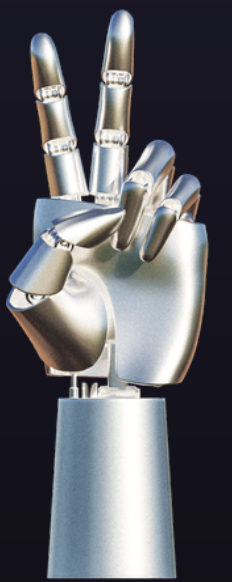
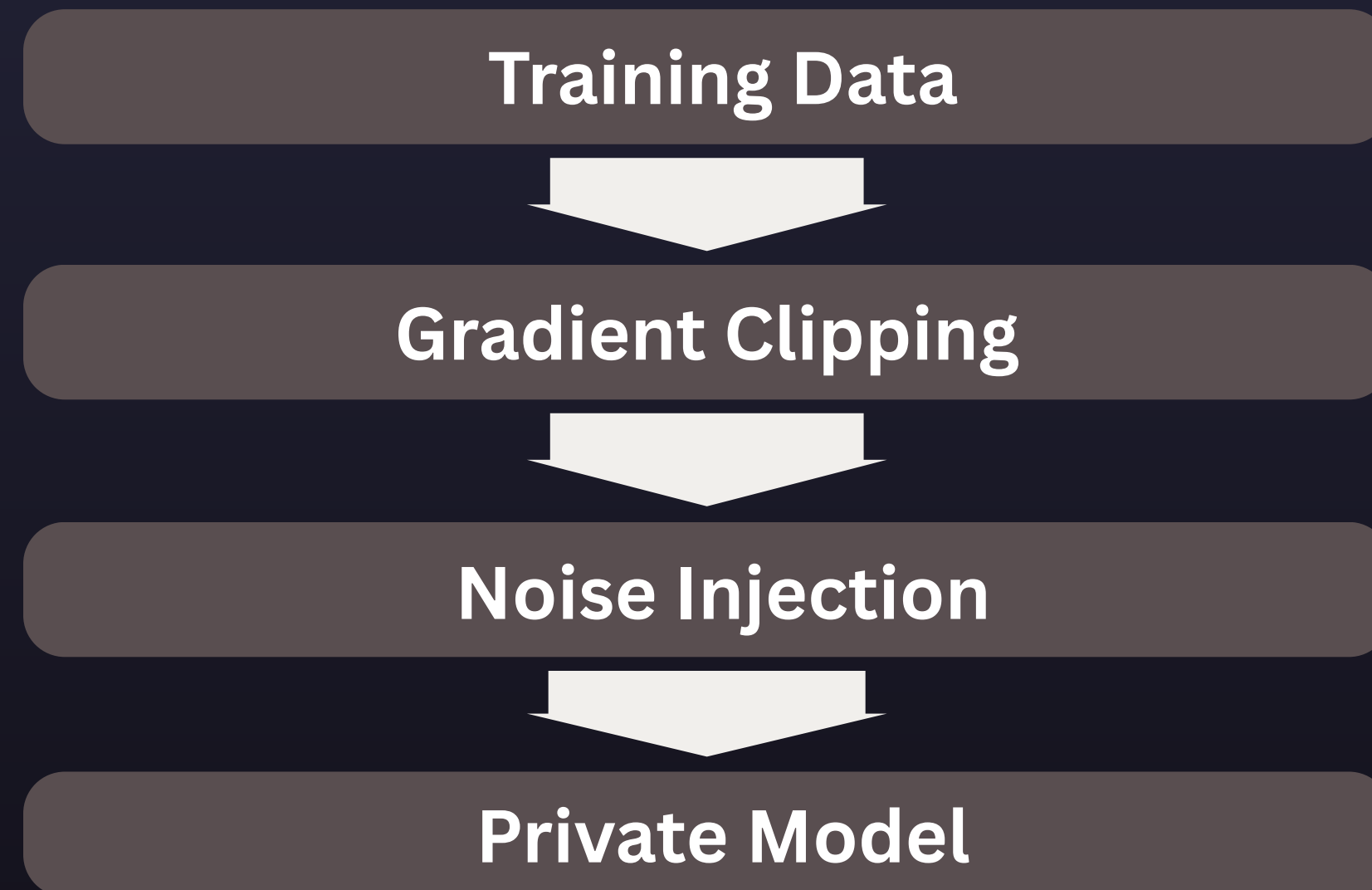


OUTCOME



DIFFERENTIAL PRIVACY

PRIVACY-PRESERVING LEARNING



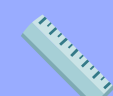
ROBUSTNESS & TRUSTWORTHY AI EVALUATION

BEYOND ACCURACY: CAN WE TRUST THE MODEL'S PREDICTIONS?



Calibration

Trustworthiness of
probability estimates



Conformal Prediction

Confidence-aware
predictions



Bayesian Uncertainty

Model knows when it is
unsure



FGSM Attack

Stress-testing model
reliability



MAJOR FINDINGS

1. Sophisticated models failed to significantly outperform baseline.
2. Most features show overlapping distributions between classes.
3. ROC-AUC remained near random guessing.
4. Dataset contains very weak predictive signals.
5. Relationship outcomes may depend on factors absent from the dataset.

**MODEL COMPLEXITY
CANNOT COMPENSATE
FOR THE ABSENCE OF
PREDICTIVE SIGNAL.**

CONCLUSION

Achievements:

- ✓ Built complete ML pipeline
- ✓ Evaluated 14 ML models
- ✓ Applied Explainable AI
- ✓ Applied Causal Inference

Key Scientific Finding:

Despite extensive modelling, relationship outcomes remain highly unpredictable using the available features.

Future Work:

- Real-world dating datasets
- Text analysis of conversations
- Graph neural networks
- Sequential behavioural modelling

CONTRIBUTIONS :

- ✓ **END-TO-END ML PIPELINE**
- ✓ **EXPLAINABLE AI**
- ✓ **CAUSAL INFERENCE**
- ✓ **DIFFERENTIAL PRIVACY**
- ✓ **DEPLOYMENT-READY SYSTEM**